

ARENA First Nations Clean Energy Resilience Fund

1. Program Overview

- **Funding Agency:** Australian Renewable Energy Agency (ARENA)
 - **Program Name:** First Nations Community Energy Resilience Fund (FNCERF)
 - **Goal:** To deploy solar PV, battery energy storage systems (BESS), and microgrids in remote First Nations townships to guarantee energy security, reduce reliance on diesel generators, and ensure continuous operation of essential community services (clinics, water pumps, emergency comms).
 - **Funding Value:** Grants between \$500,000 and \$3,000,000.
 - **Co-Funding Requirement:** Applicants must provide a minimum 15% co-contribution (cash or in-kind, including state/territory civil infrastructure allocations).
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2. Technical and Operational Requirements

- **System Focus:** Off-grid solar-battery-diesel hybrid microgrids, or stand-alone commercial solar array installations on essential buildings.
 - **Component Standards:** Solar panels and battery containers must be rated for remote/desert environments (temperatures up to 50°C, dust resistance, and cyclone-rated mounts where applicable).
 - **Power Quality:** Systems must integrate automatic transfer switches (ATS) and smart energy management software to manage grid-islanding during power outages.
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3. Written Application Criteria

Applicants must address the following criteria in detail:

Criterion 1: Energy Resilience & Critical Load Analysis (Weight: 35%)

- *Prompt:* Detail the critical electrical load requirements of your target facility (e.g., vaccine refrigeration, diagnostic equipment, water pump power). Explain how the proposed solar and battery storage capacity is sized to support these critical loads during a continuous 48-hour mains power failure.

- *Key Information Required:* Average daily energy demand (kWh), critical load profile, and system sizing specifications (kW solar, kWh battery).

Criterion 2: Technology Selection & Operational Reliability (Weight: 35%)

- *Prompt:* Justify your selection of technology components (inverters, battery chemistry, microgrid controller). Explain your strategy for system monitoring, preventative maintenance, and troubleshooting in a remote region where technician travel times are significant.
- *Key Information Required:* Starlink telemetry monitoring, local maintenance training plan, component warranty details.

Criterion 3: Partnership, Budget & Financial Co-investment (Weight: 30%)

- *Prompt:* Detail your project budget, including all freight, shipping, and installation costs. Confirm the sources of your 15% co-contribution. Explain how this clean energy installation will lower the organisation's long-term utility costs and improve clinical operational continuity.
- *Key Information Required:* Budget tables, proof of co-funding reserves, 10-year operational cost saving forecast.